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NEET · CHEMISTRY · CHAPTER-WISE PYQS

# Nuclear Chemistry

NEET Chemistry · Previous Year Questions with Answer Key

| 19 | 14 | 1989 – 2022 | 7 | English |
|---|---|---|---|---|
| QUESTIONS | PAPERS COVERED | YEARS | OUT OF SYLLABUS | LANGUAGE |

INSIDE THIS PDF

| | |
|---|---|
| MCQ (Single Correct Answer) | 12 questions |
| Out of syllabus | 7 questions |
| Answer key | at the end |

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## MCQ (Single Correct Answer) 12 questions

**Q1.** NEET 2022 Phase 2

The half life of a first order reaction is 2000 years. If the concentration after 8000 years is 0.02 M, then the initial concentration was :

(A) 0.04 M (B) 0.16 M (C) 0.32 M (D) 0.08 M

Open solution

**Q2.** AIPMT 2005

A nuclide of an alkaline earth metal undergoes radioactive decay by emission of three $\alpha$-particles in succession. The group of the periodic table to which the resulting daughter element would belong is

(A) Gr.13 (B) Gr. 17 (C) Gr. 14 (D) Gr. 16

Open solution

**Q3.** AIPMT 2004

The radioactive isotope $^{60}_{27}$Co which is used in the treatment of cancer can be made by (n, p) reaction. For this reaction the target nucleus is

(A) $^{59}_{28}$ Ni (B) $^{59}_{27}$ Co (C) $^{60}_{28}$ Ni (D) $^{60}_{27}$ Co

Open solution

**Q4.** AIPMT 2003

The radioistope, tritium ($^{3}_{1}$H) has a half-life of 12.3 years. If the initial amount of tritium is 32 mg, how many milligrams of it would remain after 49.2 years?

(A) 1 mg (B) 2 mg (C) 4 mg (D) 8 mg

Open solution

**Q5.** AIPMT 2002

$^{235}_{92}$U, nucleus absorbs a neutron and disintegrates into $^{139}_{54}$Xe, $^{94}_{38}$Sr and $x$ so, what will be product $x$?

(A) 3-neutrons (B) 2-neutrons (C) $\alpha$-particle (D) $\beta$-particle

Open solution

**Q6.** AIPMT 2001

A human body required 0.01 m activity of radioactive substance after 24 hours. Half life of radioactive substance is 6 hours. Then inhection of maximum activity of radioactive substance that can be injected is

(A) 0.08 (B) 0.04 (C) 0.16 (D) 0.32

Open solution

**Q7.** AIPMT 2001

If a $^{a}_{b}X$ species emits first a positron, then two $\alpha$ and two $\beta$ particles and in the last, one $\alpha$, is also emitted and gets converted to $^{c}_{d}$Y species. So correct relation is

(A) c = a $-$ 12, d = b $-$ 5 (B) c = a $-$ 5, d = b $-$ 1

(C) c = a $-$ 6, d = b $-$ 0 (D) c = a $-$ 4, d = b $-$ 2

Open solution

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**Q8.** AIPMT 2000

A 300 grams radioactive sample at initial half life is 3 hours. After 18 hours remaining quantity

(A) 4.68 gram (B) 2.34 gram (C) 3.34 gram (D) 9.37 gram

Open solution

**Q9.** AIPMT 1999

Sulphur-35 ($34.96903\ \text{amu}$) emits a $\beta$-particle but no $\gamma$-ray. The product is chlorine-35 ($34.96885\ \text{amu}$). The maximum energy emitted by the $\beta$-particle is

(A) $16.758\ \text{MeV}$ (B) $1.6758\ \text{MeV}$ (C) $0.16758\ \text{MeV}$ (D) $0.016758\ \text{MeV}$

Open solution

**Q10.** AIPMT 1999

In the following radioactive decay, ${}_{92}\text{X}^{232} \longrightarrow {}_{89}\text{Y}^{220}$, how many $\alpha$ and $\beta$ particles are ejected from X to Y?

(A) $3\alpha$ and $2\beta$ (B) $5\alpha$ and $3\beta$ (C) $3\alpha$ and $3\beta$ (D) $5\alpha$ and $5\beta$

Open solution

**Q11.** AIPMT 1999

The half-life of ${}_{6}\text{C}^{14}$, ($\lambda = 2.31 \times 10^{-4}$ per year) is

(A) $2 \times 10^2$ year (B) $3 \times 10^3$ year (C) $3.3 \times 10^4$ year (D) $4 \times 10^3$ year

Open solution

**Q12.** AIPMT 1998

Number of neutrons in a parent nucleus $X$, which gives ${}_{7}\text{N}^{14}$ nucleus after two successive $\beta$-emissions would be

(A) 9 (B) 6 (C) 7 (D) 8

Open solution

## Out of syllabus 7 questions

*These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.*

**Q13.** AIPMT 1997 · MCQ (Single Correct Answer)

Carbon-14 dating method is based on the fact that

(A) ratio of carbon-14 and carbon-12 is constant

(B) carbon-14 is the same in all objects

(C) carbon-14 is highly insoluble

(D) all of these.

Open solution

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**Q14.** AIPMT 1997 · MCQ (Single Correct Answer)

$^{235}_{92}\mathrm{U} + n \rightarrow$ fission product + neutron $+ 3.20 \times 10^{-11}$ J. The energy undergoes fission is

(A) $8.21 \times 10^{7}$ kJ (B) $6.55 \times 10^{6}$ kJ (C) $12.75 \times 10^{8}$ kJ (D) $18.60 \times 10^{9}$ kJ

Open solution

**Q15.** AIPMT 1996 · MCQ (Single Correct Answer)

After the emission of one $\alpha$-particle followed by one $\beta$-particle from the atom of $^{238}_{92}X$, the number of neutrons in the atom will be

(A) 144 (B) 143 (C) 148 (D) 146

Open solution

**Q16.** AIPMT 1995 · MCQ (Single Correct Answer)

Half-life for radioactive $^{14}\mathrm{C}$ is 5760 years. In how many years 200 mg of $^{14}\mathrm{C}$ will be reduced to 25 mg?

(A) 17280 years (B) 23040 years (C) 5760 years (D) 11520 years

Open solution

**Q17.** AIPMT 1994 · MCQ (Single Correct Answer)

In a radioactive decay, an emitted electron comes from

(A) outermost orbit of the atom

(B) orbit having principal quantum number one

(C) nucleus of the atom

(D) inner orbital of the atom.

Open solution

**Q18.** AIPMT 1989 · MCQ (Single Correct Answer)

The age of most ancient geological formations is estimated by

(A) potassium - argon method (B) carbon - 14 dating method

(C) radium - silicon method (D) uranium - lead method.

Open solution

**Q19.** AIPMT 1989 · MCQ (Single Correct Answer)

Emission of an alpha particle leads to a

(A) decrease of 2 units in the charge of the atom

(B) increase of 2 units in the mass of the atom

(C) decrease of 2 units in the mass of the atom

(D) increase of 4 units in the mass of the atom.

Open solution

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# Answer Key

## MCQ (Single Correct Answer)

| | | | | |
|---|---|---|---|---|
| 1. **C** | 2. **C** | 3. **C** | 4. **B** | 5. **A** |
| 6. **C** | 7. **A** | 8. **A** | 9. **C** | 10. **C** |
| 11. **B** | 12. **A** | | | |

## Out of syllabus

| | | | | |
|---|---|---|---|---|
| 13. **A** | 14. **A** | 15. **B** | 16. **A** | 17. **C** |
| 18. **D** | 19. **A** | | | |

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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